

Supplementary Material for “The Wrong Unit of Uncertainty: Simultaneous Conformal Bands for Repeated Cross-National Surveys”

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This online supplement collects secondary material referenced from the main manuscript. All content, numbers, and figures are reproduced unchanged; only figure numbers carry an “S” prefix.

S1 Proofs

This section gives full statements and proofs of the results of the main text, in the main-manuscript numbering. Each result is paired with a machine-checked contract test in the replication package that verifies its guarantee numerically. Where the source argument is a sketch or rests on an estimated constant, we say so.

Common setup. We observe K exchangeable source countries $c = 1, \dots, K$ plus a target $K+1$. Each country c has a latent attitude curve F_c (a CDF over a grid $t_1 < \dots < t_T$) and a transport center μ (a leave-one-out mean or fitted center), with plug-in error curve $E_c(t) = \hat{\theta}_t(P_c) - \theta_t(P_c)$. The latent transport score is $R_c = \max_{r,t} |F_c(t) - \mu(t)|/\sigma(t)$, the country-level (exchangeable-unit) score. F_c is not observed; a complex survey returns $\tilde{F}_c = F_c + S_c$ with design error S_c , inducing the contaminated score $\tilde{R}_c = R_c + \xi_c$ with $\mathbb{E}[\xi_c | R_c] = 0$ and $\text{Var}(\xi_c | \cdot) = v_c^2$, where v_c^2 is estimated by a stratified-PSU / replicate-weight design bootstrap. Write $s_{\text{plug}}^2 = s_R^2 + \overline{v^2}$ and the design-to-total SD ratio $\rho = \sqrt{\overline{v^2}}/s_{\text{plug}} \in [0, 1)$. Two targets are kept distinct throughout: the survey-estimate score $\tilde{R}_{K+1}$ (T1) and the latent score R_{K+1} (T2), the object the political claim concerns.

S1.1 Theorem 1 (non-identification and necessity of design information)

Statement. Let the observed score be $\tilde{R} = R + \xi$ with $R \perp \xi$ and ξ mean-zero. (i) The law of R —hence its $(1 - \alpha)$ quantile—is not identified from the law of $\tilde{R}$ without restrictions on the law of ξ . (ii) Among bands measurable with respect to the observed scores only, none is valid for the latent target at level $1 - \alpha$ for all admissible design laws while being strictly narrower than the plug-in band whose radius is the $(1 - \alpha)$ conformal quantile of $|\tilde{R}|$.

Proof. (i) Convolution is not invertible without knowing the noise law. Take $\mathcal{L}(\tilde{R}) = N(0, \sigma^2)$. Then $(R \sim N(0, \sigma^2), \xi \equiv 0)$ and, for any $0 < \tau^2 < \sigma^2$, $(R \sim N(0, \sigma^2 - \tau^2), \xi \sim N(0, \tau^2))$ both have $\xi \perp R$, ξ mean-zero, and induce the same $\mathcal{L}(\tilde{R})$; their latent $(1 - \alpha)$ quantiles $z_{1-\alpha}\sigma$ and $z_{1-\alpha}\sqrt{\sigma^2 - \tau^2}$ differ. The same construction applies to any $\mathcal{L}(\tilde{R})$ that is a nontrivial convolution. (ii) Under the member $\xi \equiv 0$ the latent target equals $\tilde{R}_{K+1}$, which is exchangeable with $\{\tilde{R}_c\}_{c \leq K}$; the conformal rank identity gives $\Pr(\tilde{R}_{K+1} \leq \hat{q}) = \lceil (1 - \alpha)(K+1) \rceil / (K+1)$ at $\hat{q}$ the corresponding

order statistic, so any radius below the $\lceil(1-\alpha)(K+1)\rceil$ -th order statistic of $\{|\tilde{R}_c|\}$ has latent coverage $< 1 - \alpha$ on this member. A band narrower than the plug-in on a positive-measure event therefore undercovers the latent target on the $\xi \equiv 0$ law it cannot distinguish from the contaminated one. In the degenerate case $\xi \equiv 0$ the plug-in is itself the tightest valid band, so no uniform improvement exists. Thus strict efficiency is attainable only by shrinking the class of admissible laws—i.e. by supplying the law of ξ (equivalently the v_c), which is exactly the design-bootstrap information. $\square$

S1.2 Theorem 2 (oracle validity)

Statement. With the countries exchangeable and the design law known: (a) the clustered band on $\{\tilde{R}_c\}$ covers the survey-estimate target T1 exactly, $\Pr(\tilde{R}_{K+1} \leq \hat{q}) \geq 1 - \alpha$, using exchangeability only; (b) under a conditionally symmetric design law it is conservative for the latent target T2, $\Pr(R_{K+1} \leq \hat{q}) \geq 1 - \alpha$, and the deconvolution band with the true target scale has width a factor $\sqrt{1 - \rho^2} + o(1)$ of the clustered band.

Proof. (a) By exchangeability of $\{\tilde{R}_c\}_{c=1}^{K+1}$ the rank of $\tilde{R}_{K+1}$ is uniform on $\{1, \dots, K+1\}$; with $\hat{q}$ the $\lceil(1-\alpha)(K+1)\rceil$ -th order statistic, $\Pr(\tilde{R}_{K+1} \leq \hat{q}) = \lceil(1-\alpha)(K+1)\rceil/(K+1) \geq 1 - \alpha$, finite-sample and distribution-free, using no assumption on the design-noise law. (b) With ξ conditionally symmetric and independent of R , $\tilde{R} = R + \xi$ is a mean-preserving spread of R : for the upper functional $\{R \leq \hat{q}\}$ the symmetric convolution moves mass outward symmetrically, so the $(1 - \alpha)$ quantile of $\tilde{R}$ dominates that of R and a radius calibrated on $\tilde{R}$ over-covers R , giving $\Pr(R_{K+1} \leq \hat{q}) \geq 1 - \alpha - \varepsilon_{\text{sym}}$ with $\varepsilon_{\text{sym}} = 0$ under exact symmetry (a total-variation bound in the noisy-label sense supplies ε_{sym} when symmetry is only approximate). The oracle deconvolution scale satisfies $s_T^2 = s^2(1 - \rho^2)$ by the law of total variance $\text{Var}(\tilde{R}) = \text{Var}(R) + \mathbb{E}[v^2]$ under conditional mean-zero noise, giving the width factor $\sqrt{1 - \rho^2} + o(1)$. $\square$

S1.3 Theorem 3 (exact finite- K validity of the deployed base band)

Statement. With the unstudentized cluster score $R_c = \max_t |E_c(t)|$ and the countries exchangeable, for *every* K the band $\hat{B} = \text{center} \pm \hat{q}$ with $\hat{q}$ the $\lceil(1-\alpha)(K+1)\rceil$ -th order statistic of $\{R_c\}$ satisfies $\Pr(F_{\text{new}} \in \hat{B}) \geq \lceil(1-\alpha)(K+1)\rceil/(K+1) \geq 1 - \alpha$, distribution-free.

Proof. The curves $E_1, \dots, E_{K+1}$ are i.i.d. (populations i.i.d., the same estimation rule for each), hence exchangeable elements of $\mathbb{R}^T$, and by continuity the scores $\{R_c\}$ are a.s. distinct. Therefore the rank of R_{K+1} is uniform on $\{1, \dots, K+1\}$, and with $\hat{q} = R_{(m)}$, $m = \lceil(1-\alpha)(K+1)\rceil$, $\Pr(R_{K+1} \leq \hat{q}) = m/(K+1) \geq 1 - \alpha$. The simultaneous coverage event is pointwise equivalent to the scalar event: $F_{\text{new}}(t) \in \hat{B}(t) \forall t \iff |E_{K+1}(t)| \leq \hat{q} \forall t \iff \max_t |E_{K+1}(t)| \leq \hat{q} \iff R_{K+1} \leq \hat{q}$. Hence simultaneous coverage is $\geq \lceil(1-\alpha)(K+1)\rceil/(K+1)$, distribution-free. Intersecting the band with $[0, 1]$ and applying the isotonic tightening $L(t) = \text{clip}_{[0,1]}(\max_{t' \leq t} \text{lo}(t'))$, $U(t) = \text{clip}_{[0,1]}(\min_{t' \geq t} \text{hi}(t'))$ only shrink the band while still containing the monotone $[0, 1]$ -valued truth, preserving validity. The split-modulation studentized variant (modulation computed on a disjoint fold) is exact by the identical argument conditional on the modulation fold. $\square$

Modulation self-inclusion deficit (proof sketch; estimated constant). If instead one studentizes by an in-sample modulation $\hat{s}(t) = \text{SD}_c E_c(t)$ computed from the calibration curves themselves, the unobserved target does not share this denominator, which breaks exchangeability. A first-order expansion of the order-statistic bias in the number g of modulation slots each country contributes to gives a coverage deficit $\approx 0.382(g/K)$; the constant 0.382 is estimated (empirical $R^2 = 0.923$),

not derived, and is flagged as such. Pooled modulation ($g = 1$) is safe; slotwise ($g = L$) undercovers by $\approx 0.382 L/K$. This $O(g/K)$ deficit is why the deployed band is unstudentized.

S1.4 Theorem 4 (estimated-law validity)

Statement. With the design law estimated from a size- B stratified-PSU bootstrap and K countries, the deconvolution band satisfies $\Pr(F_{\text{new}} \in \hat{B}) \geq 1 - \alpha - \varepsilon_{K,B}$, where $\varepsilon_{K,B} = O(K^{-1/2}) + O(B^{-1/2}) \rightarrow 0$.

Proof sketch. By the law of total variance $\text{Var}(\tilde{R}) = \text{Var}(R) + \mathbb{E}[v^2]$ under conditional mean-zero noise; subtracting a consistent estimate $\widehat{v}^2$ identifies s_R^2 , and the deconvolved scale $\hat{s}_T^2 = \max(s_{\text{plug}}^2 - \widehat{v}^2, \text{floor})$ —in the deployed finite- K -safe form, with a *lower* confidence bound subtracted so the correction cannot over-shrink—recovers the target scale. The remainder decomposes as (i) $O(B^{-1/2})$ Monte-Carlo error in $\widehat{v}^2$, (ii) $O(K^{-1/2})$ finite- K quantile and variance estimation, and (iii) a higher-order Gaussian-shape term; each $\rightarrow 0$ and none is claimed exact. When $\widehat{v}^2 \geq s_{\text{plug}}^2$ (design noise $\gtrsim$ transport signal, $\rho \gtrsim 1$) the scale hits the floor and the selector hands off to the conservative branch. This is a proof sketch: the constants in $\varepsilon_{K,B}$ are order-of-magnitude, not tight. $\square$

S1.5 Theorem 5 (safe-adaptive finite- K validity—the deployed pipeline)

Statement. Let $\hat{J} \in \{\text{PCB}, \text{deconvolution}, \text{conservative}\}$ be the target-blind safe selector. Then, for finite (K, B) , $\Pr(F_{\text{new},r}(t) \in \hat{B}(r, t) \ \forall r, t) \geq 1 - \alpha - \delta$, where $\delta = \hat{\delta}_{\text{UCB}}(D) \leq \delta_{\text{max}}$ on the event $\{\hat{J} = \text{deconvolution}\}$ and $\delta = 0$ otherwise; δ is observable and $\rightarrow 0$ as K grows.

Proof. $\hat{J}$ is $\mathcal{F}_{\text{cal}}$ -measurable (target-blind), so $\Pr(\text{cover}) = \sum_j \Pr(\hat{J} = j) \Pr(\text{cover} \mid \hat{J} = j)$. Conditioning on $\{\hat{J} = j\}$ is conditioning on a calibration- σ -field event, under which each branch retains its marginal guarantee, because the target's rank among the calibration scores is unaffected by an $\mathcal{F}_{\text{cal}}$ -measurable choice. PCB and conservative over-cover the latent target (Theorem 2 (a)/(b)), giving conditional coverage $\geq 1 - \alpha$; deconvolution is invoked only on $\{\hat{\delta}_{\text{UCB}}(D) \leq \delta_{\text{max}}\}$, where gate B guarantees conditional coverage $\geq 1 - \alpha - \hat{\delta}_{\text{UCB}}(D)$. Each mixture term is $\geq 1 - \alpha - \delta$ with $\delta = \hat{\delta}_{\text{UCB}}(D)$ on the deconvolution event and $\delta = 0$ elsewhere; summing over the partition gives $\Pr(\text{cover}) \geq 1 - \alpha - \delta$. The remainder δ is a computable function of (D, K, B) and the noise-law error, hence observable at deployment, and $\delta \rightarrow 0$ as K grows (the region $\{D \leq \tau_D\}$ expands to the full deconvolution regime). $\square$

S1.6 Proposition 1 (survey-scale unreachability)

Statement. The deployed selector rides deconvolution only if a *need* gate $\hat{\rho}_{\text{LCB}} > \rho_0$ and a *reliability* gate $D \leq \tau_D$ both hold, with $D = \max_t \widehat{\text{SE}}(\hat{s}_T^2) / \hat{s}_T^2$. Two distribution-free facts bound these: (a) $\rho = \sqrt{\widehat{v}^2} / s_{\text{plug}} \in [0, 1)$ with $s_{\text{plug}}^2 = s_R^2 + \widehat{v}^2$; and (b) $D \geq \sqrt{2/(K-1)}$, so the reliability gate requires $K \geq 1 + 2/\tau_D^2$, i.e. $K \geq 94$ at the frozen $\tau_D = 0.147$.

Proof. (a) $\rho^2 = \widehat{v}^2 / (s_R^2 + \widehat{v}^2) \in [0, 1)$, so the need gate can fire only when the design noise $\widehat{v}^2$ is an appreciable fraction of the between-population signal s_R^2 . (b) The standard error of a variance estimated from K clusters is $\widehat{\text{SE}}(\hat{s}_T^2) = \sqrt{2s_{\text{plug}}^4 / (K-1) + (\text{SD}_K(v^2) / \sqrt{K})^2}$; since $\hat{s}_T \leq s_{\text{plug}}$ for every t ,

$$D = \max_t \frac{\widehat{\text{SE}}(\hat{s}_T^2)}{\hat{s}_T^2} \geq \frac{\sqrt{2s_{\text{plug}}^4 / (K-1)}}{s_{\text{plug}}^2} = \sqrt{\frac{2}{K-1}},$$

a distribution-free floor. Hence $D \leq \tau_D$ forces $\sqrt{2/(K-1)} \leq \tau_D$, i.e. $K \geq 1 + 2/\tau_D^2$. With the frozen gate-B constants $\hat{\delta}_{\text{UCB}}(D) = a + bD$, $a = 0.0061$, $b = 0.0943$, $\delta_{\text{max}} = 0.02$, the threshold is $\tau_D = (\delta_{\text{max}} - a)/b = 0.147$, giving $K \geq 1 + 2/0.147^2 \approx 94$. Repeated cross-national surveys have $K \leq 33$ (ESS) and ~ 20 – 40 (LAPOP, Afrobarometer, Eurobarometer), so the gate-B feasible region is empty at survey scale regardless of ρ : even if the need gate were passed, deconvolution stays blocked. $\square$

Remark 1. The $D \geq \sqrt{2/(K-1)}$ floor is fundamental—it is the sampling error of a variance estimated from K clusters, distribution-free, and gives $\geq 25\%$ relative standard error on the deconvolved scale at $K \approx 30$. The exact threshold $K \approx 94$ depends on the frozen gate-B constants (conservatively calibrated on the development grid); a less conservative δ_{max} would lower it, but the calibration-free statement is the floor itself. The barrier is specific to the *cross-national* application: many-unit settings (subnational areas, schools, clinics) routinely have $K \geq 94$, where the reliability gate can open.

S1.7 Efficiency results AW-1–AW-3 and the open oracle inequality (AW-4)

Write the three branch half-widths at level $1 - \alpha$ as $W_P = q_P s$ (PCB), $W_D = q_D s \sqrt{1 - \rho^2}$ (deconvolution), and $W_C = q_C s$ (conservative).

AW-1 (low- ρ reduction, proven). As $\rho \rightarrow 0$, $\hat{s}_T = s \sqrt{1 - \rho^2} \rightarrow s$ and the deconvolution scores converge pointwise (at fixed K) to the PCB scores, so $q_D/q_P \rightarrow 1$ by continuity of the finite-sample quantile in the studentization, and

$$W_D/W_P = (q_D/q_P) \sqrt{1 - \rho^2} = 1 - \frac{1}{2}\rho^2 + O(\rho^4)$$

by Taylor expansion. Below ρ_0 the selector returns PCB exactly, so $W_{\text{adaptive}} = W_P$.

AW-2 (conservative dominance, conditional). For $\rho \geq \rho_0$ with stable deconvolution, $W_D/W_C = (q_D/q_C) \sqrt{1 - \rho^2}$. The worst-case scores $(|E| + zv)/s$ stochastically dominate $|E|/s$, so $q_C \geq q_P$ and to first order $q_C \approx q_P(1 + z\kappa\rho)$ with $\kappa \in (0, 1]$ the correlation of the $|E|$ and v orderings across the calibration set. Hence $W_D/W_C \approx \sqrt{1 - \rho^2}/(1 + z\kappa\rho) < 1$ for all $\rho > 0$, strictly and decreasing in ρ : the conservative fallback is a genuine upper band, never wider than the Bonferroni envelope and at least as wide as PCB, and deconvolution strictly narrows it whenever design noise is present. (Conditional on the stated first-order expansion of q_C .)

AW-3 (target-blind validity, proven). $\hat{J}$ is measurable with respect to $\mathcal{F}_{\text{cal}}$ and each branch is marginally valid for its target layer; since $\hat{J} \perp \text{target}$ given $\mathcal{F}_{\text{cal}}$, $\Pr(\text{target} \in B_{\hat{J}}) = \sum_j \Pr(\hat{J} = j) \Pr(\text{target} \in B_j \mid \hat{J} = j) \geq 1 - \alpha - \max_j \varepsilon_j$, so selection cannot inflate miscoverage.

AW-4 (excess-width oracle inequality, open). Let $W^* = \min(W_P, W_D, W_C)$. Conjecture: $W_{\text{adaptive}} \leq W^* + r_{K,B}$ with $r_{K,B} = O(K^{-1/2}) + O(B^{-1/2})$ under Lipschitz width-in- ρ , the boundary zone where the target-blind ρ_0 disagrees with the oracle crossover having width $O(K^{-1/2})$ because $\hat{\rho}$ is $\sqrt{K}$ -consistent. This remains open: (i) $\rho_0 = 0.47$ is fixed from the development grid rather than derived as the oracle crossover $W_D = W_C$, and (ii) the Lipschitz constant of width-in- ρ is not established. It is stated as a candidate with a proof sketch, not a theorem.

S2 The finite- K mechanism and the development grid

Development: why the third gate is needed. On a Gaussian grid with known truth ($K \in \{30, 60, 120, 240\}$, ρ up to 1.8), the selector transitions from PCB to deconvolution at ρ_0 and to conservative at large ρ ; but a naive deconvolution that omits gate B *undercovers* at small K —coverage falls to 0.75 at $K = 30$ in the transition zone—because the target scale is estimated from

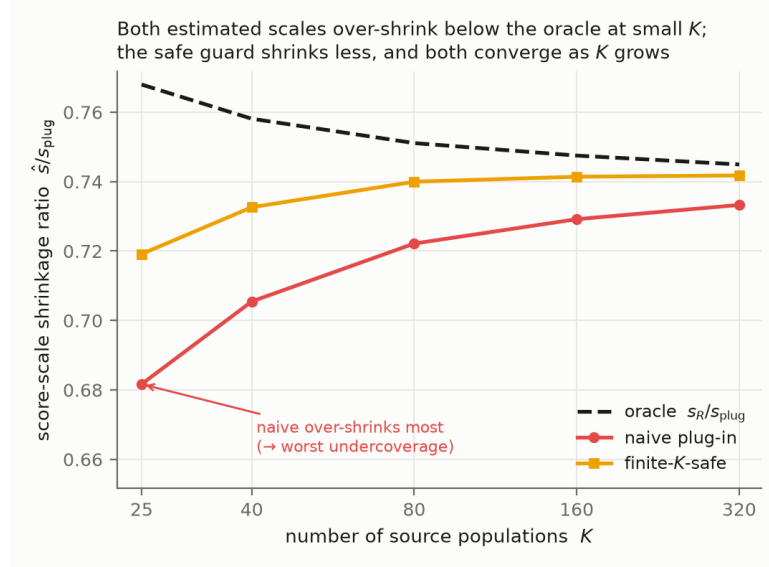


Figure S1: Score-scale shrinkage ratio $\hat{s}/s_{\text{plug}}$ vs K (Gaussian illustration, $\rho=0.9$, $L=8$). Naive and finite- K -safe deconvolution both over-shrink below the oracle at small K —the source of the finite- K undercoverage—with the safe guard shrinking less; both converge as K grows.

few countries (Figure S3). Adding the finite- K -safe scale lifts this to 0.82, and the reliability gate B lifts the deployed worst cell to 0.862 while abstaining to conservative wherever the remainder cannot be bounded. These four transition cells at 0.862–0.878 are the development evidence that *motivates* the safety gate and *calibrates* $\hat{\delta}_{\text{UCB}}$; they are not reused below.

The finite- K mechanism, isolated. A controlled Gaussian illustration (design SD = transport SD, $\rho=0.9$; `fig_theory_illustration.py`) makes the source of the undercoverage explicit and shows why the deployed rule is a selector rather than a deconvolver. Both the naive and the finite- K -safe deconvolved score scales lie *below* the oracle s_R/s_{plug} at small K (Figure S1): subtracting a design-noise variance estimated from few countries over-shrinks the scale, and the resulting band is too narrow. The safe guard shrinks less, and both converge to the oracle as $K \rightarrow \infty$. The coverage consequence is severe and grows with the trajectory length L that the band must hold simultaneously (Figure S2): naive deconvolution falls to 0.43 at $K=25$, $L=16$, and the finite- K -safe scale improves it but does not by itself reach nominal at small K . The deployed adaptive selector detects that the remainder cannot be bounded and abstains to the conservative branch, remaining valid throughout at a width cost—the mechanism behind the $K=320$ -only activation in the confirmatory holdout in the main text.

S3 Extended robustness of the remainder bound

Cell-by-cell resolution of the base-method deficit. On the holdout the studentized band’s worst deployed cell was 0.843 (weak cross-round dependence at $K=25$), a finite- K self-inclusion deficit worst at small K and long trajectories. After switching to the unstudentized clustered band, a preregistered rerun on a fresh grid ($K \in \{15, \dots, 60\}$, ten DGPs) confirms the fix: the worst cell rises from 0.802 (S2, $K=15$) to 0.893, every cell meets its exact floor, and the width cost is 1.02–1.05 $\times$ on the real ESS transport bands (trust-CDF errors are $\approx$ homoskedastic across the low-trust

Both naive and finite- K -safe deconvolution undercover at small K (worse as L grows); the deployed selector abstains and stays valid

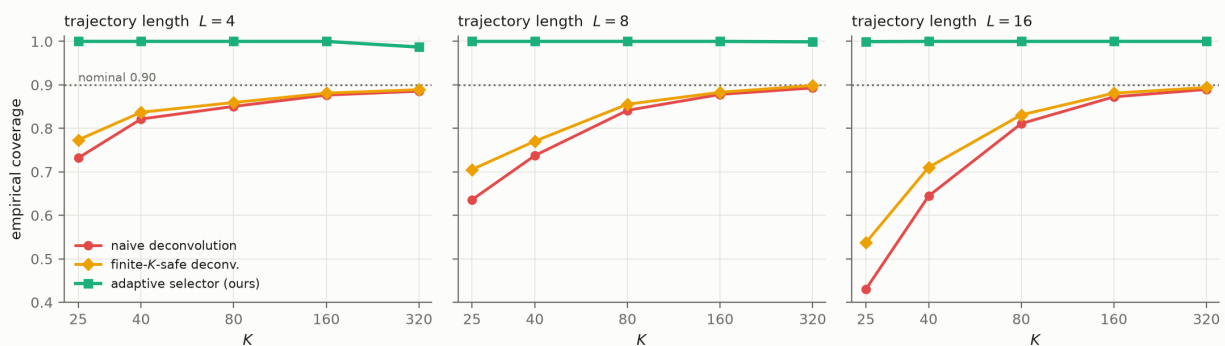


Figure S2: Latent-target coverage of the deconvolution branches vs K , faceted by trajectory length L (Gaussian illustration, $\rho=0.9$; nominal 0.90 dotted). Naive deconvolution undercovers severely at small K and worse as L grows; the finite- K -safe scale improves but does not itself reach nominal at small K . The deployed adaptive selector abstains to conservative and stays valid throughout.

core, so the constant-width band is competitive; the only cells where it pays more than 10% are synthetic heavy-tailed curves at $K=15$, where the studentized alternative is itself invalid).

Extreme-tail behavior of the remainder bound. Beyond the base-method deficit resolved by deploying the unstudentized clustered band, the holdout exposed a second, extreme-tail limitation. The Gaussian-calibrated remainder bound $\hat{\delta}_{\text{UCB}}$ is not robust at the extreme: in 77 of 360,000 draws (0.02%), on strongly-dependent or heavy-tailed DGPs at $K=320$ and $\rho \geq 0.72$, draws that pass gate B undercover to ≈ 0.79 ; because the gates route 96–98% of draws in those cells to conservative, the deployed cell coverage there is nonetheless 0.95–0.996.

S4 Additional empirical robustness

Design-preserving stress test. Subsampling the real AmericasBarometer while preserving its stratified-PSU structure, the estimated ρ saturates near 0.23: since $s_{\text{plug}}^2 = s_R^2 + \bar{v}^2$, shrinking the sample inflates design noise and observed spread in lockstep. The deconvolution regime is therefore unreachable even semi-synthetically at survey scale—direct evidence for the regime characterization of the main text.

The barriers are not survey artifacts. Both gates are distribution-free, so the boundary should transfer to any conformal procedure calibrated on estimated objects (the general remark of the main text). A simulation confirms it outside the survey setting (`e29_beyond_surveys`): across Gaussian, heavy-tailed, skewed, and a non-survey model-based small-area (MRP-style) calibration, the reliability floor $D \geq \sqrt{2/(K-1)}$ holds in every case (minimum $D/\text{floor} \geq 1.06$), and the gate opens only at $K \approx 130$ –150—if anything a stronger barrier than the floor-implied $K \geq 94$. The one substantive difference is instructive: when a small area’s posterior noise is comparable to the between-area signal ($\hat{\rho}_{\text{LCB}} = 0.54$ at a noise-to-signal ratio of 0.5), the *need* gate does fire, unlike any cross-national survey. A many-area MRP application with appreciable posterior uncertainty is therefore the setting where the deconvolution could actually activate—the positive-result frontier the survey-scale impossibility leaves open.

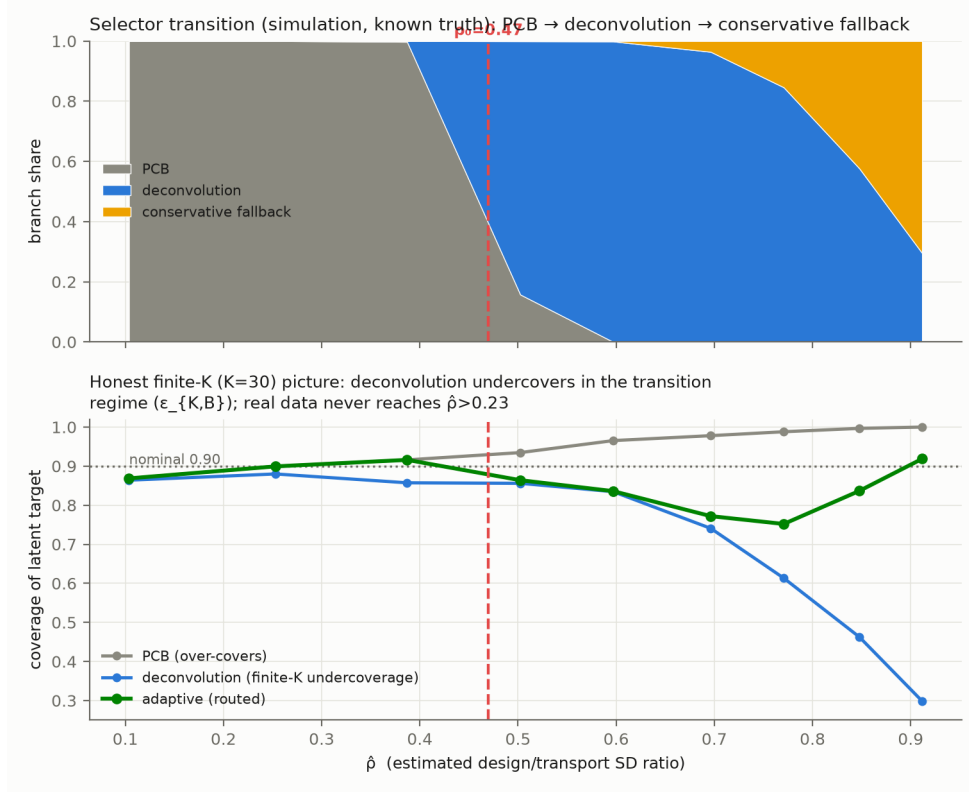


Figure S3: Development grid (Gaussian, known truth). The selector transitions PCB → deconvolution → conservative as ρ grows; naive deconvolution undercovers at small K in the transition zone—the finite- K remainder the safety gate controls.

The AmericasBarometer supplies the design effect the ESS lacks. The ESS releases design weights but limited cluster identifiers; the AmericasBarometer releases full stratified-PSU structure, letting us compute a proper design bootstrap. The design effect is real: the ratio of the proper stratified-PSU standard deviation to a naive independent-resampling one has median ≈ 1.13 – 1.20 and reaches 1.9 in the most clustered country-years. The proper band captures clustering variance the naive bootstrap misses. Yet certification is coarse-robust: the design effect moves band *width* but rarely flips the binary decline decision, because the decision is dominated by the between-country transport spread, not the within-country design noise—the same low- ρ fact, seen from the certification side.

Robustness across age groups (ESS). A preregistered age-group analysis (preregistered in the replication package) reruns the same certification within fixed age bands, changing nothing else. The “only Greece” result is not an artifact of pooling ages: it holds within the older (50+) group for both outcomes and within the middle (30–49) group for trust in parliament, so the persistent decline is concentrated in the older and middle adult population rather than created by averaging. Youth (18–29) certify *zero* persistent declines, but we read this as an underpowered null rather than youth stability: only 17 of 30 countries clear the preregistered minimum-sample gate for a single youth band, and the smaller, noisier youth cells yield wider design-aware bands that certify less—the small- K scope condition of the simulation appearing on real data. The analysis thus supports the headline while adding an honest caveat, and it is emphatically *not* a story of youth-led democratic

disaffection.

Youth deconsolidation on the World Values Survey. The youth claim in the Foa–Mounk reanalysis is only half-supported: the young show *more* persistent decline in rating democracy essential (7 vs. 3 countries) but *not* in openness to strongman or army rule (0–1 vs. 4), against the “youth embrace authoritarianism” reading.

References